

CTU Training Solutions
NET731 – Networking 3A
Week 2 Practical: Cloud Network Design

13 – 19 March 2025 | Total Marks: 10

Module Code	NET731
Module Name	Networking 3A
Week	2 (13 – 19 March 2025)
Topic	Cloud Network Design
Total Marks	10
Submission	GitHub repository link submitted via the student portal
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Student Number	20240429

Microsoft Azure Upgrade

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Home > Resource Manager | Resource groups

Create a resource group ...

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure subscription 1

Resource group name * ⓘ TechBridge-RG

Region * ⓘ (Africa) South Africa North

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Home > Network foundation | Virtual networks

Create virtual network

Basics

Security

Address space

Tags

Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.

[Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure subscription 1

Resource group *

TechBridge-RG

[Create new](#)

Instance details

Virtual network name *

TechBridge-VNet

Region * ⓘ

(Africa) South Africa North

[Deploy to an Azure Extended Zone](#)

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Create virtual network

Basics Security **Address space** Tags Review + create

Virtual network address space

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

☐ Allocate using IP address pools. [Learn more](#)

+ Add a subnet

10.0.0.0/16 [Delete address space](#)

10.0.0.0 /16

10.0.0.0 - 10.0.255.255 65,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-

Add IPv4 address space

Advertised address prefixes

+ Add address prefix

Advertised gateway prefixes let you control route advertisements by specifying which IP address ranges Azure gateways advertise to on premises networks, instead of advertising all virtual network and spoke CIDRs.

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Create virtual network ...

Validation passed

Basics Security Address space Tags Review + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource Group	TechBridge-RG
Name	TechBridge-VNet
Region	South Africa North

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

Address space

Address space	10.0.0.0/16 (65,536 addresses)
Subnet	default (10.0.0.0/24) (256 addresses)
- Private subnet	Enabled

Tags

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TechBridge-VNet | Subnets

Virtual network

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Monitoring

Automation

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to
☐	default	10.0.0.0/24	-	251	-
☐	Public-Sub...	10.0.1.0/24	-	251	-
☐	Internal-Su...	10.0.2.0/24	-	251	-

Showing 3 subnets

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https://portal.azure.com/#@gigglesquiver1kcloud.onmi...

Create network security group ...

Basics Tags Review + create

Project details

Subscription *

Azure subscription 1

Resource group *

TechBridge-RG

[Create new](#)

Instance details

Name *

echBridge-NSG

Region *

South Africa North

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Search resources, services, and docs (G+I)

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Search subnets

Name	Address range	Virtual network
Internal-Subnet	10.0.2.0/24	TechBridge-VNet
Public-Subnet	10.0.1.0/24	TechBridge-VNet

Give feedback

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